

Friedrichs Aharonov–Bohm Dynamics: Domain and Complete Hankel Spectrum

HCS-C383 theorem and reproducibility package

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Abstract

We fix the Friedrichs magnetic Laplacian on the punctured Euclidean plane, with one flux and one physical time convention. Angular Fourier decomposition reduces the closed form to radial Bessel operators of orders $|m - \beta|$. The Hankel Plancherel theorem gives an onto spectral transform in every channel and consequently a purely absolutely continuous spectrum $[0, \infty)$. The exceptional low-order channels retain the regular branch and exclude the singular branch; zero flux recovers the ordinary free Laplacian because a point has zero planar Sobolev capacity. This first manuscript closes the operator domain and the complete spectral representation. It is a reconstruction and audit of classical source theory, not a claim of a new Aharonov–Bohm formula.

中文摘要

本文为穿孔欧氏平面上的阿哈罗诺夫玻姆磁性动力学固定弗里德里希斯实现、磁通符号与物理时间。角向傅里叶分解把闭二次型化为各阶径向贝塞尔算子的正交直和，阶数由角动量与磁通之差的绝对值决定。汉克尔变换的普朗歇雷与反演定理给出每个通道的满射谱表示，因此完整谱为非负实轴上的纯绝对连续谱，没有束缚态或奇异连续谱。低阶通道必须通过二次型选择正规分支；零阶允许常数行为而排除对数奇性，不能误用逐点零边界条件。零磁通时，点的二维索伯列夫容量为零，穿孔不会改变自由平面拉普拉斯的闭型域。本轮完成算子定义、边界约定与全通道谱，属于经典理论的可审查重建。

Keywords: Aharonov–Bohm Hamiltonian; Friedrichs extension; Hankel transform; Bessel spectrum; punctured plane

关键词：阿哈罗诺夫玻姆算子；弗里德里希斯扩张；汉克尔变换；贝塞尔谱；穿孔平面

1 One operator and one boundary condition

On $\mathcal{H} = L^2(\mathbb{R}^2, r dr d\theta)$, set $h = 2M = 1$ and freeze the nonnegative form, initially on smooth functions compactly supported away from zero,

$$q_\beta[f] = \int_0^\infty \int_0^{2\pi} \left(|\partial_r f|^2 + \frac{|(-i\partial_\theta - \beta)f|^2}{r^2} \right) r d\theta dr. \quad (1)$$

Its closed Friedrichs extension defines the self-adjoint H_β . The main flux representative is $0 \leq \beta < 1$; integer translates will only be used for gauge comparisons. No spin, contact interaction, alternate self-adjoint extension, finite-radius flux tube, or confining field enters. This choice is material: an unqualified singular differential expression does not fix its quantum dynamics.

Let $e_m(\theta) = (2\pi)^{-1/2} e^{im\theta}$ and $\nu_m = |m - \beta|$. Angular Parseval gives the orthogonal form sum

$$q_\beta = \bigoplus_{m \in \mathbb{Z}} q_{\nu_m}, \quad q_\nu[u] = \int_0^\infty \left(|u'|^2 + \frac{\nu^2}{r^2} |u|^2 \right) r dr. \quad (2)$$

Finite angular sums and smooth compact radial approximants form a core of this sum: Fourier truncation converges in both norm and form energy, and the radial closure then approximates each of finitely many components.

Theorem 1 (Domain and complete spectral representation). *The Friedrichs operator decomposes as*

$$H_\beta = \bigoplus_{m \in \mathbb{Z}} h_{\nu_m}, \quad h_\nu = -\partial_r^2 - r^{-1} \partial_r + \nu^2 r^{-2}. \quad (3)$$

The order- ν Hankel map

$$(\mathcal{J}_\nu u)(k) = \int_0^\infty J_\nu(kr) u(r) r dr \quad (4)$$

is unitary from $L^2(r dr)$ onto $L^2(k dk)$ and transforms h_ν to multiplication by k^2 . In particular,

$$D(h_\nu) = \{u : k^2 \int_\nu u \in L^2(k dk)\}. \quad (5)$$

The full operator has purely absolutely continuous spectrum $[0, \infty)$, with no point or singular continuous spectrum.

Proof. The form-core argument proves (3). For $0 < \nu < 1$, the two zero-energy branches are r^ν and $r^{-\nu}$. The latter has divergent form energy and its coefficient is removed in the Friedrichs realization. For $\nu = 0$, the logarithmic branch is excluded but the regular constant branch is allowed. Orders $\nu \geq 1$ are limit point at zero, and infinity is limit point in every channel. Thus these are precisely the regular Bessel realizations diagonalized by the Hankel Plancherel and inversion theorem [3]. The transform is onto and gives (5); it is not a finite list of formal eigenfunctions. Multiplication by k^2 under the non-atomic measure $k dk$ has the claimed spectral types, and its countable orthogonal sum preserves them. $\square$

Lemma 1 (The zero-flux puncture). H_0 is the ordinary free plane Laplacian.

Proof. For $0 < \epsilon < 1$, take a radial cutoff equal to zero at $r \leq \epsilon^2$, equal to one at $r \geq \epsilon$, and linear in $\log r$ between those radii. Its Dirichlet energy is $2\pi/|\log \epsilon|$. Multiplying a fixed smooth function by this cutoff changes its L^2 norm and Dirichlet energy by terms tending to zero. Hence the puncture has zero H^1 capacity and the closed zero-flux form equals the full-plane free form. $\square$

The original physical effect belongs to Aharonov and Bohm [1]; the rigorous spectral and scattering treatment for general extension parameters is reviewed by Pankrashkin and Richard [2]. Their magnetic-potential sign corresponds to orders $|m + \alpha|$. Angular reflection, or $\alpha = -\beta$, translates to (3). The present formulas use one sign throughout. Reconstructing and testing this classical theory is the purpose of this package; classical formula discovery is not claimed.

References

- [1] Y. Aharonov and D. Bohm, “Significance of Electromagnetic Potentials in the Quantum Theory,” *Physical Review* 115 (1959), 485–491, doi:10.1103/PhysRev.115.485.
- [2] K. Pankrashkin and S. Richard, “Spectral and scattering theory for the Aharonov–Bohm operators,” preprint (2009), arXiv:0911.4715.
- [3] NIST Digital Library of Mathematical Functions, Section 10.17 and Section 10.22, Hankel transforms and Weber’s exponential integral, accessed 5 September 2026.

round zero domain and complete Hankel spectrum